ABBA6 Implicit

One symmetric projection around a Yoshida composition

Explicit palindromic operations

All ABBA and BM4 presets inherit `PalindromicMethod`. For signed $d = ch$, the elementary operations are

$$A_d(u,v,t) = (u + df(t,v), v, t), \qquad B_d(u,v,t) = (u, v + df(t,u), t), \qquad T_d(u,v,t) = (u,v,t+d).$$

Only T advances time. A coefficient $c = 1/2$ means duration $h/2$; there is no additional half inside A or B. The optional exact C flow rotates the copy difference and requires an independent coupling object. Both BM4 presets omit C.

In chronological order, standard ABBA2 is $A_{h/2}, B_{h/2}, T_{h/2}, T_{h/2}, B_{h/2}, A_{h/2}$. The distinct symmetric variant $T_{h/2}, A_{h/2}, B_{h/2}, B_{h/2}, A_{h/2}, T_{h/2}$ evaluates every spatial shear at the midpoint time. Both exhibit second order on the tested nonautonomous rotation. Arbitrary palindromes do not inherit fourth or sixth order, and an arithmetic projection need not be reversible or symplectic.

`Composition.from_half` reflects a supplied half sequence. The generic `PalindromicImplicit` and `PalindromicMidpoint` methods select one exterior Hairer projection or a final arithmetic mean. The standard signed ABBA/BM4 recipes, grouped observations and passive momentum normalization remain unchanged. NumPy and JAX execute the same explicit operations and differentiate their retained shear inputs. See `docs/models/extended/simulation/palindromic-operations.md`.

1 Unprojected seven-pair composition

Let $S_{q,t}$ denote the symmetric second-order ABBA map on two spatial copies, with signed duration q and start time t. Set

$$a = 0.78451361047755726382, \qquad b = 0.23557321335935813368,$$
$$c = -1.17767998417887100695, \qquad d = 1.31518632068391121889,$$

and $(w_1, \dots, w_7) = (a, b, c, d, c, b, a)$. For a trial multiplier μ, the complete base map is evaluated through

$$Y^{(0)} = Ez_n + N\mu, \qquad t_j = t_n + h\sum_{\ell<j} w_\ell, \qquad Y^{(j)} = S_{w_j h, t_j}(Y^{(j-1)}), \qquad \Psi_{h,t_n}(Y^{(0)}) = Y^{(7)}.$$

Both copies remain independent between pairs. Each pair applies an adjoint and a direct map with weights $w_j/2$, giving fourteen stages in total. The weights satisfy, up to round-off,

$$\sum_j w_j = 1, \qquad \sum_j w_j^3 = 0, \qquad \sum_j w_j^5 = 0.$$

These moments and palindromy are useful checks, but are not sufficient on their own to prove order six: the Yoshida solution also satisfies the mixed commutator order condition for a symmetric second-order base.

2 One outer symmetric projection

For $m = 2N_p$ physical coordinates, with N_p independent particles, define $Ez = (z, z)$, $G = [I, -I]$, $N = G^{\mathsf{T}}$ and $P = \frac{1}{2}[I, I]$. One complete step solves

$$r(\mu) = G\Psi_{h,t_n}(Ez_n + N\mu) + 2\mu = 0,$$
$$z_{n+1} = P\big(\Psi_{h,t_n}(Ez_n + N\mu) + N\mu\big).$$

The reduced formulation has m unknowns. The simultaneous formulation solves for (Y_+, μ) using

$$Y_+ - N\mu - \Psi_{h,t_n}(Ez_n + N\mu) = 0, \qquad GY_+ = 0.$$

Both formulations define the same local map and permit Newton or Broyden. There is one multiplier and one nonlinear solve per complete step. This is exactly the ABBA4 architecture with a different unprojected recipe. It intentionally replaces the historical composition of seven projected ABBA2 maps; previously saved trajectories and costs belong to that former map.

3 Signed times, order and geometry

The third and fifth pair durations are negative. Internal times can leave $[t_n, t_n + h]$, reaching approximately $t_n - 0.15759h$ and $t_n + 1.15759h$; the dynamics must be defined throughout the signed path. Adjoint stages evaluate at their starting clock, direct stages at their ending clock.

For a smooth field, the Yoshida base has order six on doubled space. Its exact flow preserves the diagonal, so the local diagonal defect is $O(h^7)$. At $h = 0$, $D_\mu r = 4I$; the implicit function theorem therefore gives the local multiplier $\mu = O(h^7)$. The outer projection preserves order six. The symmetric root construction is reversible and, for the canonical Hamiltonian splitting, symplectic on the physical diagonal. These statements concern the local exact root. Finite stopping tolerances, round-off and nonsmooth fields can limit the observed order and geometry.

4 Accepted traces and passive energy

Only the accepted root contributes to passive energy. The 28 signed shear sources give

$$\Delta\kappa = \frac{1}{2}\sum_j s_j[-\partial_t H(t_j, z_j)].$$

Time and momentum do not enter the nonlinear equation. The ABBA6 observation contains one multiplier, complete-step work metrics and seven unprojected pair snapshots. Work histories have shape $(n_{\text{steps}}, 1)$ rather than $(n_{\text{steps}}, 7)$. Differentiation composes the unprojected tangents first, then differentiates the single outer root.

5 Implementation correspondence and verification

The public class is in `src/methods/extended/abba.py`; it inherits the same initialization, advance and placement validation as ABBA4. Recipes and weights live in `src/methods/extended/core/composition.py`. The path is `advance`, `solve_projection`, `compose`, then explicit $A/B/T/C$ operations. Common Newton/Broyden services, particle-block tangents and energy quadrature are shared with BM4. Public method names remain unchanged; no forwarding modules preserve retired internal paths. Event conversion lives in `extended/observations.py`.

Tests check the one-solve contract, fourteen signed stages, pair continuity, nonautonomous sixth-order convergence, both residual formulations with both solvers, reversibility, analytic tangents and symplecticity. The nonlinear convergence test uses an independently tighter DOP853 reference. Current architecture and diagrams are documented in `docs/models/extended/simulation/extended-simulation-architecture.md`.

6 Four state formulations and passive energy tracking

For one planar particle the supported internal representations are

physical	$z = (x, y)$	$\mathbb{R}^2$,
physical with energy	(z, t, κ)	$\mathbb{R}^4$,
duplicated	$(u, v), \quad u, v \in \mathbb{R}^2$	$\mathbb{R}^4$,
duplicated with energy	(u, v, t, κ)	$\mathbb{R}^6$.

Classical methods use the physical rows; ABBA and BM4 use the duplicated rows. `track_energy` selects the energy variant. For N planar particles the respective dimensions are $2N$, $4N$, $4N$, and $6N$: one time entry and one momentum per particle. All time entries equal the integration time. Classical full-cyclotron runs instead use $4N$ or $6N$ entries. Accepted spatial copies coincide; only one physical copy is exported. Every coordinate block contains N values in component-major order. The formulation validates particle times and aligns output round-off; the integrator does not interpret coordinate indices.

The normalized passive balance is

$$\dot{\kappa} = -\partial_t H(t, z), \qquad \kappa(t_0) = 0, \qquad \Delta K(t) = H(t, z(t)) + \kappa(t) - H(t_0, z_0).$$

Splitting accumulators from both copies are divided by two. The physical Hamiltonian may vary with time; ΔK measures its computed balance. Tracking leaves physical trajectories, nonlinear stopping rules and adaptive acceptance unchanged. Dynamics must supply the momentum derivative, including zero for an autonomous Hamiltonian.

Hairer's constraint is solely $u - v = 0$: the reduced multiplier has $2N$ entries and ABBA's simultaneous spatial solve has $6N$ unknowns. Neither t nor κ is projected. The former eight-component projection is retired; `state_extension='fully_extended'` raises migration guidance.

Dynamics, formulation, method and integration remain separate. The shared state classes own layouts, diagonal embedding and history extraction in

```
src/formulations/state.py
```

Each integration initializes a fresh method instance. Observers receive physical states; energy histories are in `Solution.diagnostics`. The `extended_time` history has shape (N, n_{samples}). Fixed-step shadow samples leave main states and counters unchanged. Adaptive energy quadrature uses accepted dense output; its backend state and Jacobian remain physical.

7 Optional JAX execution

An optional JAX driver executes the same numerical map on one selected CPU or GPU, with particle operations batched inside a compiled sequential time loop. It reuses the canonical coefficients and prepared potential splines. Physical-state tolerance scaling, signed stage times, passive energy weights and independent shortened output maps retain their definitions above. Applicable Newton systems use independent particle blocks; an explicitly selected Broyden iteration retains its global secant matrix. Nonlinear iteration limits and failure checks remain active on the device.

Both execution routes use double precision and return the same physical solution layout. Floating-point reassociation can produce round-off differences without changing the designed order or geometric properties. First-call compilation is separate from warmed integration timings, which include transfers and output construction. The sibling runtime document and shared execution contract specify supported callbacks and exact device scope.